Practice Quiz 08B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 08

Testing and repairing a rule on congruence classes.

For a positive integer m, write $\mathbf{Z}_m$ for the set of congruence classes modulo m. Recall that $[x]_m = [y]_m$ if and only if $x - y = mk$ for some integer k.

1. (a) Show that the proposed rule

$$h : \mathbf{Z}_6 \longrightarrow \mathbf{Z}_4, \qquad h([x]_6) = [x]_4$$

is not well defined. Exhibit two representatives of the same input class that give different output classes.

Solution The integers 0 and 6 represent the same class in $\mathbf{Z}_6$, since $6-0 = 6\cdot 1$. But the proposed outputs are $[0]_4$ and $[6]_4 = [2]_4$, which are different: 4 does not divide 6. Therefore the output depends on the representative, so the rule is not well defined. ∎

(b) Prove that changing the codomain gives a well-defined rule

$$g : \mathbf{Z}_6 \longrightarrow \mathbf{Z}_3, \qquad g([x]_6) = [x]_3.$$

Solution Let $x, y \in \mathbf{Z}$ and suppose $[x]_6 = [y]_6$. Then $x - y = 6k$ for some integer k. Hence $x - y = 3(2k)$, where $2k$ is an integer, so $[x]_3 = [y]_3$. Thus any two representatives of the same input class give the same output class, and g is well defined. ∎

Solutions revised using GPT-6 Astra in the Codex harness.